SkyHigh Analytics

Detailed Project Report

Project Domain: Data Analytics & SQL
Technology: MySQL | Dataset: 3 Relational Tables
Scope: 1,000 Passengers | 50 Flights | 5,000 Bookings

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1. Project Overview

This report presents a comprehensive SQLbased analysis of an airline booking dataset, conducted under the project title Analytics. The project aims to convert raw transactional airline data into actionable business intelligence that can help airline companies make informed decisions about passenger targeting, route optimization, pricing strategy, and revenue generation.

1.1 Project Title

1.2 Objective

The primary objective of this project is to analyze passenger demographics, flight operations, and booking revenue patterns using structured query language (SQL) to:

- Understand who is flying and from where
- Identify the most popular airlines, routes, and destinations
- Analyze booking behavior and seasonal patterns
- Evaluate airline revenue performance
- Provide actionable business recommendations based on data

1.3 Business Problem

The airline industry generates enormous volumes of data from passenger bookings, flight operations, and ticketing systems. Without proper data analysis, airlines face several critical challenges:

- Inability to identify highvalue customer segments for targeted marketing
- Lack of insight into which routes and destinations generate the most demand
- Difficulty in understanding pricing strategies across seat classes
- No clear view of seasonal booking trends to optimize fleet and staffing
- Missed opportunities to reward loyal passengers and reduce churn

1.4 Scope of Analysis

The analysis is structured into four distinct phases:

Phase	Focus Area
Phase 1	Passenger Demographics & Behavior
Phase 2	Flight & Route Operations
Phase 3	Booking & Revenue Analysis
Phase 4	Advanced CrossTable Analysis (Joins)

2. Dataset Description

The project utilizes three interrelated relational database tables stored in a MySQL database named **SkyHigh Analytics**. Together, these tables contain comprehensive information about passengers, available flights, and booking transactions.

2.1 Database Schema

Table Name	Primary Key	Key Columns	Record Count
Passengers	passenger_id	name, age, gender, country	~1,000
Flights	flight_id	airline, source_city, destination_city, distance_km	50
Bookings	booking_id	passenger_id (FK), flight_id (FK), ticket_price, seat_class, booking_date	~5,000

2.2 Table Details

2.2.1 Passengers Table

Stores demographic information for each passenger. This table is the foundation of passengerlevel analysis.

- **– Primary key, unique identifier for each passenger**
- **– Full name of the passenger**
- **– Age of the passenger in years**
- **– Gender (Male / Female)**
- **– Country of origin / residence**

2.2.2 Flights Table

Contains operational information about each available flight route.

- **– Primary key, unique identifier for each flight**
- **– Name of the operating airline**
- **– City of departure**
- **– City of arrival**
- **– Flight distance in kilometres**

2.2.3 Bookings Table

The central transactional table connecting passengers to flights. It was modified during the project to fix a data type issue the booking_date column was converted from TEXT to DATE for accurate datebased calculations.

- **– Primary key, unique booking record**
- **– Foreign key referencing Passengers**
- **– Foreign key referencing Flights**
- **– Price paid for the ticket (in INR ₹)**
- **– Business or Economy**
- **– Date of booking (converted to DATE type)**

2.3 Database Relationships

The three tables follow a standard relational model:

- One passenger can make multiple bookings (1:N Passengers to Bookings)
- One flight can have multiple bookings (1:N Flights to Bookings)
- The Bookings table serves as a bridge/junction table between Passengers and Flights

This design enables complex JOIN queries to analyze crossdimensional data such as airline revenue by passenger country, or demographic spending patterns by class.

3. Phase 1 – Passenger Insights

This phase focuses on understanding the demographic composition of the airline's passenger base. Insights here directly inform marketing segmentation and customer experience strategies.

3.1 Gender Distribution

Query: Total count of Male vs. Female passengers.

Gender	Passenger Count	Percentage
Female	511	51.1%
Male	489	48.9%

Finding: Female passengers slightly outnumber male passengers, comprising 51.1% of the total dataset. While the split is nearly even, this small skew may inform gendertargeted promotions or onboard service personalization.

3.2 Top 5 Countries by Passenger Count

Query: Identifying which countries contribute the most passengers.

Rank	Country	Passenger Count
1	United Kingdom (UK)	127
2	France	107
3	UAE	104
4	USA	101
5	Australia	100

Finding: The United Kingdom leads with 127 passengers, followed closely by France and UAE. This indicates that European and Gulfbased markets are the primary customer base. Airlines can prioritize marketing spend and route planning to target these geographies.

3.3 Average Passenger Age

Query: Average age of all passengers in the dataset.

Metric	Value
Average Age	44.9 years (≈ 45 years)
Interpretation	Predominantly middleaged adult travellers

Finding: The average passenger age of approximately 45 years suggests a mature, workingage customer base. This demographic typically has disposable income, values comfort, and may prefer Business class travel.

3.4 Age Group Segmentation

Query: Classifying passengers into Teens (≤19), Adults (20–50), and Seniors (50+).

Age Group	Age Range	Count	Percentage
Adult	20 – 50 years	561	56.1%
Senior	50+ years	414	41.4%
Teens	≤ 19 years	25	2.5%

Finding: Adults (20–50) form the overwhelming majority at 56.1%, followed by Seniors at 41.4%. Teenagers represent only 2.5% of passengers. This data strongly indicates the airline's core demographic is workingage adults and retirees both prime candidates for loyalty programs and premium service offerings.

3.5 Business Class Popularity by Country

Query: Countries where Business Class bookings are highest.

Country	Business Class Bookings
United Kingdom	304
France	~260
UAE	~248

Finding: The UK not only leads in total passenger count but also in Business Class bookings (304). This confirms the UK market as the most premiumoriented market in the dataset. Targeted Business class promotions in the UK could yield the highest ROI.

4. Phase 2 – Flight & Route Operations

This phase analyzes flightlevel data to understand which airlines dominate the market, which destinations attract the most traffic, and what the average operational parameters look like.

4.1 Airline Popularity

Query: Airlines operating the most flights in the dataset.

Rank	Airline	Number of Flights
1	Lufthansa	9
2	Air India	9
3	British Airways	8
4	Qatar Airways	7
5	Singapore Airlines	6

Finding: Lufthansa and Air India are tied as the most frequently operating airlines with 9 flights each. British Airways follows closely with 8. These three carriers form the backbone of the flight network in this dataset.

4.2 Top Destination Cities

Query: The 3 most frequently visited destination cities.

Rank	Destination City	Number of Flights
1	Doha (Qatar)	9
2	Frankfurt (Germany)	8
3	Singapore	7

Finding: Doha leads as the most frequent destination, reflecting Qatar Airways' hub and Gulf tourism demand. Frankfurt represents the European business travel hub, while Singapore anchors the Southeast Asian market. These three cities should be prioritized for route expansion and capacity planning.

4.3 Average Flight Distance

Query: Average distance of all flights in the dataset.

Metric	Value
Average Flight Distance	4,189.3 km
Classification	Medium to LongHaul Flights
Implication	Premium pricing and comfort services are expected by passengers

Finding: An average distance of 4,189 km places these flights in the mediumtolonghaul category. Passengers on such routes are more likely to value comfort, meal service, and class upgrades reinforcing the case for Business class expansion.

5. Phase 3 – Booking & Revenue Analysis

This phase dives into the financial and temporal dimensions of the booking data answering questions about total revenue, ticket pricing, seasonal patterns, and booking day preferences.

5.1 Total Revenue

Query: Total sum of all ticket prices in the dataset.

Metric	Value
Total Revenue	₹11,47,49,866 (approx. ₹11.47 Crore)
Data Coverage	All 5,000 booking records across all seat classes

5.2 Seat Class Price Comparison

Query: Average ticket price for Business Class vs. Economy Class.

Seat Class	Average Ticket Price (₹)	Revenue Multiplier
Business Class	₹36,905	4.15×
Economy Class	₹8,905	1×

Finding: Business class tickets are priced over 4 times higher than Economy. Despite likely having fewer seats, Business class contributes disproportionately to airline revenue. Expanding Business class capacity or introducing a Premium Economy tier could significantly boost margins.

5.3 Monthly Booking Volume

Query: Which month saw the highest number of bookings?

Rank	Month	Bookings
1 (Peak)	March	468
2	January	453
3	May	432
11	October	409

12 (Lowest)	November	374

Finding: March is the peak booking month with 468 bookings, likely driven by spring holiday travel and business conferences. November is the quietest month at 374 bookings. Airlines should align pricing, staffing, and promotions with this seasonal curve.

5.4 Booking Day Preferences

Query: Which day of the week has the highest booking volume?

Day	Bookings	Rank
Sunday	746	1st (Peak)
Tuesday	727	2nd
Monday	726	3rd
Friday	704	4th
Wednesday	703	5th
Thursday	701	6th
Saturday	693	7th (Lowest)

Finding: Sunday sees the most bookings (746), which is counterintuitive as it is a day of rest. This suggests passengers plan their travel during leisure time on weekends. Saturday has the fewest bookings. Airlines could introduce flash sales on Sunday evenings to capitalize on this peak engagement window.

5.5 Ticket Price Range

Query: Minimum and maximum ticket prices in the dataset.

Metric	Value
Minimum Ticket Price	₹3,005
Maximum Ticket Price	₹69,989
Price Spread	₹66,984

Finding: The wide range between ₹3,005 and ₹69,989 reflects both the EconomyBusiness class divide and variance across routes of different lengths. Dynamic pricing strategies can leverage this spread to optimize yield management.

6. Phase 4 – Advanced CrossData Analysis

This phase uses SQL JOIN operations across all three tables to derive composite insights that cannot be obtained from a single table alone. These queries reveal revenue attribution, passenger loyalty, and demographic preferences.

6.1 Airline Revenue Rankings

Query: Total revenue generated by each airline (JOIN: Bookings + Flights).

Rank	Airline	Total Revenue (₹)	Revenue in Crores
1	Lufthansa	₹2,16,01,867	₹2.16 Cr
2	Air India	₹2,05,03,348	₹2.05 Cr
3	British Airways	₹1,70,43,404	₹1.70 Cr
4	Qatar Airways	₹1,64,12,658	₹1.64 Cr
5	Singapore Airlines	₹1,41,47,807	₹1.41 Cr

Finding: Lufthansa leads total revenue generation at ₹2.16 Crore, with Air India close behind at ₹2.05 Crore. Despite their smaller network in this dataset, both airlines have maximized revenue through premium pricing and higher booking volumes. British Airways comes third at ₹1.70 Crore.

6.2 Passenger Loyalty

Query: Passengers who have made more than 3 bookings (JOIN: Passengers + Bookings).

Finding: Several passengers were identified as having booked 4 or more flights. These frequent flyers represent a highvalue segment. Offering a dedicated loyalty program with mile accumulation, priority boarding, and class upgrade benefits could significantly improve retention for these passengers.

6.3 Most Popular Airline Among Indian Passengers

Query: Airline preference filtered by passengers from India (JOIN across all 3 tables).

Rank	Airline	Bookings by Indian Passengers
1	Air India	85
2	Lufthansa	64
3	British Airways	64
4	Qatar Airways	~58

Finding: Air India dominates among Indian passengers with 85 bookings a natural outcome given its national carrier status and extensive Indian route network. Lufthansa and British Airways are also popular, indicating that Indian passengers frequently travel to European destinations. Air India should leverage this brand loyalty through tailored inflight services, regional cuisine, and language support.

7. Key Findings Summary

#	Finding	Impact
1	Adults (20–50) form 56% of all passengers	Core demographic for product design and marketing
2	UK leads in passenger count (127) and Business class (304 bookings)	Priority market for premium services
3	Business class yields 4× higher average revenue than Economy	High revenue leverage from premium expansion
4	Lufthansa is the #1 revenue generator at ₹2.16 Crore	Benchmark for airline performance
5	March has peak bookings (468); November lowest (374)	Seasonal capacity planning opportunity
6	Sunday sees peak booking volume (746)	Ideal window for flash sales and promotions
7	Air India dominates among Indian passengers (85 bookings)	Strong brand loyalty to leverage in India
8	Average flight distance: 4,189 km (mediumtolong haul)	Passengers expect premium inflight experience
9	Total dataset revenue: ₹11.47 Crore	Strong commercial viability of routes
10	Doha, Frankfurt, Singapore are top 3 destinations	Highpriority routes for capacity investment

8. Business Recommendations

Based on the findings across all four phases of analysis, the following strategic recommendations are proposed for airline management teams:

8.1 Expand HighDemand Routes

Doha, Frankfurt, and Singapore consistently appear as the top 3 destination cities. Airlines should increase flight frequency on these routes and consider expanding to secondary cities within those regions (e.g., Dusseldorf, Dubai, Kuala Lumpur) to capture adjacent demand.

8.2 Scale Business Class Capacity

With Business Class generating over 4× the revenue per seat compared to Economy, there is a strong financial case for:

- Increasing Business Class seat allocation on longhaul routes
- Introducing a Premium Economy tier to capture midmarket passengers
- Partnering with corporate travel desks in the UK and France the top Business class markets

8.3 Target European & Gulf Markets

UK, France, and UAE collectively account for 338 of the 1,000 passengers the top three source countries. Airlines should:

- Run geotargeted digital ad campaigns in these regions
- Offer loyalty miles bonuses for passengers booking from these countries
- Partner with local travel agencies in London, Paris, and Dubai

8.4 Launch Sunday Promotional Campaigns

With Sunday being the peak booking day (746 bookings), airlines can:

- Run timelimited 'Sunday Saver' deals to further increase conversions
- Send personalized email/push notifications on Sunday mornings with targeted offers
- Use Sunday data to test A/B pricing experiments with minimal business disruption

8.5 Build a Passenger Loyalty Program

Frequent flyers booking 3+ flights are prime loyalty candidates. Recommendations:

- Create a tiered points system (Silver, Gold, Platinum) based on annual bookings
- Offer free upgrades to Business Class for loyalty members on select routes

- Partner with hotels and car rental companies in top destinations for bundled rewards

8.6 Seasonal Capacity Planning

March is the peak booking month while November is the slowest. Airlines should:

- Preposition additional aircraft and crew for March operations
- Offer earlybird discounts for November bookings to smooth demand
- Analyze whether November dip is driven by route seasonality or economic factors

8.7 Air India IndiaSpecific Strategy

Air India's dominance among Indian passengers (85 bookings) provides a foundation for:

- Launching Indiaexclusive deals on Air India routes
- Investing in regional language support and cuisine on Indiaorigin flights
- Targeting India's growing middle class with Economy upgrade promotions

9. SQL Methodology Notes

This project was executed entirely using MySQL. Below are notes on the key SQL techniques applied across phases:

9.1 Data Preparation

- Converted booking_date column from TEXT to DATE using ALTER TABLE ... MODIFY to enable date functions
- Used DESC command to inspect table schemas before querying
- Verified referential integrity between Passengers, Flights, and Bookings tables

9.2 Key SQL Techniques Used

Technique	Purpose	Example Query Context
GROUP BY + COUNT	Aggregate passenger counts by gender, country	Gender distribution, top countries
AVG()	Compute average values	Avg age, avg distance, avg ticket price
CASE WHEN	Conditional segmentation	Age group classification (Teens/Adult/Senior)
ORDER BY + LIMIT	TopN ranking queries	Top 5 countries, top 3 destinations
MONTHNAME() / DAYNAME()	Date extraction functions	Peak month and day analysis
INNER JOIN (23 tables)	Crossdimensional analysis	Airline revenue, regional airline popularity
HAVING clause	Postaggregation filtering	Passengers with >3 bookings
FORMAT() / CONCAT()	Output formatting	Revenue display with commas, price with Rs prefix
SUM()	Total revenue calculation	Airline revenue aggregation

10. Conclusion

The Analytics project successfully demonstrates the power of SQL in transforming raw airline booking data into meaningful business intelligence. Across four analysis phases, the project delivered insights spanning passenger demographics, route operations, revenue performance, and booking behavior.

The three most impactful findings are:

- Business Class is a revenue multiplier generating 4× more revenue per ticket than Economy, making it the single highestleverage area for airline profitability
- The UK is the key market leading in both total passenger count and Business class bookings, making it the priority geography for marketing investment
- Lufthansa leads in revenue but Air India's strong performance among Indian passengers shows regional brand loyalty is a powerful asset

From a technical standpoint, this project reinforces that SQL remains one of the most effective tools for data analysis especially when dealing with structured, relational data. The combination of aggregation functions, date manipulation, conditional logic (CASE WHEN), and JOIN operations allows analysts to answer complex business questions with concise, readable queries.

Going forward, this dataset could be enriched with flight delay records, customer satisfaction scores, and ancillary revenue data to enable an even deeper layer of analysis including predictive modeling for booking demand and customer churn.

End of Report

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